

| Variable                              | Wilcoxon signed ranks test |       |        | Kruskal–Wallis one-way analysis |       |         |
|---------------------------------------|----------------------------|-------|--------|---------------------------------|-------|---------|
|                                       | W–W1                       | W–W20 | W1–W20 | W–C                             | W1–C1 | W20–C20 |
| Total fish abundance ( <i>N</i> )     | 0.462                      | 0.116 | 0.345  | 0.008                           | 0.238 | 0.669   |
| Total juvenile abundance ( <i>J</i> ) | 0.173                      | 0.138 | 0.598  | 0.055                           | 0.330 | 0.104   |
| Total adult abundance ( <i>A</i> )    | 0.249                      | 0.138 | 0.225  | 0.008                           | 0.747 | 0.104   |
| Two-spotted goby ( <i>N</i> )         | 0.028                      | 0.028 | 1      | 0.008                           | 0.469 | 0.221   |
| Two-spotted goby ( <i>J</i> )         | 0.043                      | 0.043 | 0.180  | 0.055                           | 0.648 | 0.221   |
| Two-spotted goby ( <i>A</i> )         | 0.028                      | 0.043 | 0.180  | 0.008                           | 0.280 | 0.221   |
| Black goby ( <i>N</i> )               | 0.028                      | 0.028 | 0.115  | 1                               | 0.453 | 0.108   |
| Black goby ( <i>J</i> )               | 0.042                      | 0.086 | 0.197  | 1                               | 0.103 | 0.328   |
| Black goby ( <i>A</i> )               | 0.028                      | 0.028 | 0.225  | 1                               | 0.666 | 0.328   |

**Table 2.** *P*-values from statistical comparison between the wind-turbine transects (Wilcoxon signed ranks test) and from comparisons between control and turbine transects (Kruskal–Wallis one-way analysis)  
Variables are total fish density (*N*), juveniles (*J*) and adults (*A*)

| Area                                    | Temperature (°C) | Salinity   | Depth (m) |
|-----------------------------------------|------------------|------------|-----------|
| <b>Baseline (2003 to 2005)</b>          |                  |            |           |
| WF                                      | 9.6 ± 1.0        | 10.9 ± 2.9 | 5.9–9.1   |
| SR                                      | 9.4 ± 0.9        | 9.1 ± 1.6  | 5.8–8.7   |
| NR                                      | 9.5 ± 1.3        | 15.4 ± 6.0 | 6.1–8.6   |
| <b>Operational phase (2008 to 2010)</b> |                  |            |           |
| WF                                      | 10.7 ± 1.8       | 8.4 ± 2.4  | 5.9–9.1   |
| SR                                      | 10.8 ± 1.8       | 8.3 ± 2.2  | 5.8–8.7   |
| NR                                      | 10.6 ± 1.3       | 9.6 ± 3.8  | 6.1–8.6   |
| WF_T                                    | 12.4 ± 1.6       | 8.1 ± 0.7  | 4.5–9.0   |

| Species                       | Common name         | WF   | SR | NR |
|-------------------------------|---------------------|------|----|----|
| <i>Agonus cataphractus</i>    | Hooknose            | A    | A  | A  |
| <i>Anguilla anguilla</i>      | Eel                 | BA   | BA | BA |
| <i>Centrolabrus exoletus</i>  | Rock cook           | –    | –  | A  |
| <i>Ctenolabrus rupestris</i>  | Goldsinny wrasse    | BA   | A  | BA |
| <i>Cyclopterus lumpus</i>     | Lumpfish            | BA   | B  | BA |
| <i>Gadus morhua</i>           | Cod                 | BA   | BA | BA |
| <i>Gobius niger</i>           | Black goby          | BA   | BA | BA |
| <i>Merlangius merlangus</i>   | Whiting             | A(T) | –  | –  |
| <i>Myoxocephalus scorpius</i> | Shorthorn sculpin   | BA   | BA | BA |
| <i>Platichthys flesus</i>     | Flounder            | BA   | BA | BA |
| <i>Pleuronectes limanda</i>   | Dab                 | A    | A  | BA |
| <i>Pleuronectes platessa</i>  | Plaice              | BA   | A  | BA |
| <i>Psetta maxima</i>          | Turbot              | B    | A  | –  |
| <i>Raniceps raninus</i>       | Tadpole fish        | A    | –  | –  |
| <i>Scophthalmus rhomus</i>    | Brill               | –    | A  | –  |
| <i>Solea solea</i>            | Sole                | A    | –  | A  |
| <i>Symphodus melops</i>       | Corkwing wrasse     | A    | –  | BA |
| <i>Taurulus bubalis</i>       | Longspined bullhead | BA   | BA | BA |
| <i>Zeugopterus punctatus</i>  | Topknot             | BA   | –  | –  |
| <i>Zoarces viviparus</i>      | Eelpout             | BA   | BA | BA |

|                  |                                                               |                                                                 |                                                                |                                                                   |
|------------------|---------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------|
| Survival exp. 1. | Bothnian Sea sediment<br>Bothnian Sea amphipods<br>19.7 ± 5.1 | Bothnian Sea sediment<br>Baltic proper amphipods<br>21.25 ± 5.2 | Baltic proper sediment<br>Bothnian Sea amphipods<br>18.7 ± 3.8 | Baltic proper sediment<br>Baltic proper amphipods<br>17.6 ± 9.9   |
| Survival exp. 2  | <i>Thalassiosira weissflogii</i><br>12.6 ± 9.9                | <i>Anabaena</i> sp<br>9.2 ± 10.0                                | <i>Rhodomonas salina</i><br>9.8 ± 7.1                          | <i>Thalassiosira weissflogii</i><br>and competition<br>17.0 ± 9.8 |

To each microcosm 50 adults is added and the sex ratio is expected to be 1:1 (Sundelin & Eriksson 2008, Sundelin et al., 2008). There were no significant differences in survival between treatments. Data represents mean values of 5 replicates ± SD.

| Fatty acids       | Female Baltic proper | Female Bothnian Sea |
|-------------------|----------------------|---------------------|
| 14:0              | 11.23                | 12.63               |
| 15:0              | 2.29                 | 1.23                |
| 16:0              | 37.34                | 31.63               |
| 17:0              | 1.56                 | 0.86                |
| 18:0              | 3.66                 | 2.98                |
| 19:0              | 0.46                 | 0.38                |
| 20:0              | 0.64                 | 0.86                |
| 22:0              | 0.07                 | 0.10                |
| Σ Saturated       | 57.26                | 50.68               |
| Σ Branched*       | 3.04                 | 1.86                |
| 16:1n-7           | 64.79                | 56.64               |
| 17:1n-8           | 2.52                 | 1.03                |
| 18:1n-9           | 29.20                | 40.06               |
| 18:1n-7           | 7.49                 | 5.21                |
| 20:1n-9           | 0.82                 | 0.87                |
| 20:1n-7           | 1.10                 | 0.63                |
| 22:1n-11          | 0.07                 | 0.01                |
| 22:1n-9           | 0.92                 | 0.60                |
| Σ Monounsaturated | 106.89               | 105.05              |
| 16:3n-3           | 3.76                 | 2.26                |
| 16:4n-3           | 2.25                 | 1.64                |
| 18:2n-6           | 4.61                 | 4.54                |
| 18:3n-6           | 0.05                 | 0.14                |
| 18:3n-3           | 2.41                 | 1.31                |
| 18:4n-3           | 7.52                 | 5.83                |
| 20:2n-6           | 0.53                 | 0.43                |
| 20:4n-6           | 5.05                 | 3.04                |
| 20:3n-3           | 0.24                 | 0.14                |
| 20:4n-3           | 0.71                 | 0.64                |
| 20:5n-3           | 24.44                | 22.86               |
| 22:4n-6           | 0.38                 | 0.35                |
| 22:5n-6           | 0.70                 | 0.36                |
| 22:5n-3           | 0.58                 | 0.44                |
| 22:6n-3           | 12.13                | 14.70               |
| Σ Polyunsaturated | 65.81                | 59.06               |
| Σ HUFA            | 53.76                | 49.85               |
| Σ Unsaturated     | 172.71               | 164.11              |
| Σ (n-3)           | 54.04                | 49.82               |
| Σ (n-6)           | 11.31                | 8.86                |
| (n-3)/(n-6)       | 4.78                 | 5.62                |
| DHA/EPA           | 0.50                 | 0.64                |
| Σ Total FAME      | 232.54               | 216.27              |

\* (Iso 14:0, Anteiso 14:0, Iso 16:0 and Anteiso 16:0).

in Fig. 3. Table adapted from FIEDLER *et al.* (2017).

| Trait                   | Category                | Label  | Relevance                                                                   |
|-------------------------|-------------------------|--------|-----------------------------------------------------------------------------|
| Body size               | Continuous <sup>a</sup> | –      | Growth rate, productivity, metabolism, feeding interactions                 |
| Longevity               | Very short (< 1 yr)     | vsho   | Life cycle/lifespan, productivity                                           |
|                         | Short (1–2 yrs)         | sho    |                                                                             |
|                         | Long (2–5 yrs)          | lon    |                                                                             |
|                         | Very long (5–10 yrs)    | vlon   |                                                                             |
| Reproductive frequency  | Annual episodic         | anep   | Reproduction, productivity                                                  |
|                         | Annual                  | anpr   |                                                                             |
|                         | protracted              |        |                                                                             |
| Living habit            | Semelparous             | sem    | Living environment, dispersal, foraging mode                                |
|                         | Attached                | att    |                                                                             |
|                         | Burrow dweller          | budw   |                                                                             |
|                         | Free                    | free   |                                                                             |
| Feeding position        | Tube dweller            | tub    | Food acquisition, feeding mode                                              |
|                         | Suspension feeder       | sus    |                                                                             |
|                         | Surface feeder          | surf   |                                                                             |
|                         | Sub-surface feeder      | susurf |                                                                             |
|                         | Selection feeder        | sel    |                                                                             |
| Resource capture method | Miner                   | min    | Food acquisition, complementary to <i>Feeding position</i> : summarize diet |
|                         | Cirri                   | cirr   |                                                                             |
|                         | Jawed                   | jaw    |                                                                             |
|                         | Net                     | net    |                                                                             |
|                         | Pharynx                 | phar   |                                                                             |
| Movement type           | Radula                  | rad    | Mobility, dispersal, ability to escape predation                            |
|                         | Siphon                  | siph   |                                                                             |
|                         | No movement             | nom    |                                                                             |
|                         | Swimmer                 | swim   |                                                                             |
|                         | Rafter-drifter          | raft   |                                                                             |
|                         | Crawler                 | crawl  |                                                                             |
|                         | Byssus threads          | byss   |                                                                             |
| Body design             | Tube                    | tube   | Body structure, protection against predation                                |
|                         | Burrower                | burr   |                                                                             |
|                         | Articulate              | art    |                                                                             |
|                         | Bivalved                | biv    |                                                                             |
|                         | Conical                 | con    |                                                                             |
|                         | Turbinate               | tur    |                                                                             |
|                         | Vermiform segmented     | ves    |                                                                             |
|                         | Vermiform unsegmented   | veun   |                                                                             |
| Sociability             | Solitary                | sol    | Social behaviour                                                            |
|                         | Gregarious              | greg   |                                                                             |
|                         | Aggregated              | agg    |                                                                             |

<sup>a</sup> Obtained from individual species measurements from the samples.

| Taxon                     | Average abundance (# alga <sup>-1</sup> ) |      |      | Contrib. to differences (%) |           |
|---------------------------|-------------------------------------------|------|------|-----------------------------|-----------|
|                           | 2012                                      | 2013 | 2014 | 2012–2014                   | 2013–2014 |
| <i>T. fluviatilis</i>     | 22.5                                      | 7.2  | 0.4  | 47.7                        | 21.7      |
| Hydrobia spp.             | 7.9                                       | 0.3  | 0.0  | 12.5                        | 0.9       |
| Amphipoda                 | 6.9                                       | 12.9 | 1.4  | 13.7                        | 37.6      |
| Idotea spp.               | 4.4                                       | 0.9  | 0.04 | 10.4                        | 3.0       |
| <i>M. trossulus</i>       | 4.1                                       | 4.8  | 4.1  | 10.0                        | 17.0      |
| Cerastoderma/Parvicardium | 1.6                                       | 3.0  | 1.0  | 4.3                         | 10.1      |
| <i>Jaera</i> spp.         | 0.04                                      | 1.8  | 0.0  | 0.1                         | 5.5       |

| Species                  |           | 50 – 200 m            |                        |       |       | 300 – 700 m           |                        |       |       | 1 000 – 2 000 m       |                        |       |       |
|--------------------------|-----------|-----------------------|------------------------|-------|-------|-----------------------|------------------------|-------|-------|-----------------------|------------------------|-------|-------|
|                          |           | A (I/m <sup>2</sup> ) | B (mg/m <sup>2</sup> ) | P (‰) | D (‰) | A (I/m <sup>2</sup> ) | B (mg/m <sup>2</sup> ) | P (‰) | D (‰) | A (I/m <sup>2</sup> ) | B (mg/m <sup>2</sup> ) | P (‰) | D (‰) |
| <i>Hydrobia ulvae</i>    | $\bar{x}$ | 1212                  | 932                    | 92.3  | 4.1   | 1604                  | 1468                   | 94.0  | 7.7   | 269                   | 118                    | 85.0  | 4.7   |
| PENNANT                  | $\pm$     | 319                   | 236                    | —     | 1.1   | 436                   | 445                    | —     | 1.5   | 74                    | 39                     | —     | 0.8   |
|                          | max       | 8590                  | 6088                   | —     | 37.6  | 13710                 | 12486                  | —     | 37.4  | 2725                  | 1230                   | —     | 28.6  |
| <i>Mytilus edulis</i> L. | $\bar{x}$ | 287                   | 417                    | 97.6  | 2.0   | 204                   | 485                    | 77.8  | 0.8   | 28                    | 115                    | 72.4  | 1.6   |
| (*) (**)                 | $\pm$     | 115                   | 133                    | —     | 1.2   | 99                    | 376                    | —     | 0.2   | 9                     | 52                     | —     | 0.6   |
|                          | max       | 3400                  | 2747                   | —     | 47.6  | 2607                  | 10208                  | —     | 4.3   | 150                   | 931                    | —     | 12.7  |
| <i>Mysella</i>           | $\bar{x}$ | 0                     | —                      | 0.0   | —     | 1.5                   | —                      | 7.4   | —     | 27                    | —                      | 10.0  | —     |
| <i>bidentata</i>         | $\pm$     | 0                     | —                      | —     | —     | 0.7                   | —                      | —     | —     | 18                    | —                      | —     | —     |
| MONTAGU                  | max       | 0                     | —                      | —     | —     | 30                    | —                      | —     | —     | 797                   | —                      | —     | —     |
| <i>Pholoë minuta</i>     | $\bar{x}$ | 0.6                   | 0.1                    | 3.9   | 0.0   | 1.0                   | 2                      | 30.0  | 0.1   | 14                    | 3                      | 20.0  | 0.2   |
| FABRICIUS                | $\pm$     | 0.5                   | 0.1                    | —     | 0.0   | 3                     | 1                      | —     | 0.1   | 6                     | 1                      | —     | 0.1   |
|                          | max       | 25                    | 5                      | —     | 0.1   | 100                   | 27                     | —     | 0.8   | 215                   | 82                     | —     | 1.8   |
| <i>Anaitides</i>         | $\bar{x}$ | 19                    | 38                     | 36.5  | 0.1   | 39                    | 71                     | 70.0  | 0.2   | 16                    | 7                      | 54.6  | 1.3   |
| <i>mucosa</i>            | $\pm$     | 7                     | 16                     | —     | 0.1   | 10                    | 27                     | —     | 0.0   | 5                     | 2                      | —     | 1.0   |
| BERGSTRÖM                | max       | 260                   | 647                    | —     | 1.8   | 370                   | 880                    | —     | 1.4   | 230                   | 34                     | —     | 5.5   |
| <i>Nereis diver-</i>     | $\bar{x}$ | 838                   | 2828                   | 88.5  | 3.1   | 228                   | 918                    | 96.0  | 1.6   | 36                    | 149                    | 61.8  | 1.6   |
| <i>sicolor</i>           | $\pm$     | 246                   | 733                    | —     | 0.9   | 51                    | 205                    | —     | 0.4   | 8                     | 36                     | —     | 0.4   |
| O. F. MÜLLER             | max       | 10250                 | 20323                  | —     | 28.6  | 1873                  | 3455                   | —     | 16.1  | 275                   | 1159                   | —     | 13.7  |
| <i>Nereis succinea</i>   | $\bar{x}$ | 4                     | —                      | 7.1   | 0.0   | 11                    | —                      | 25.9  | 0.3   | 0.5                   | —                      | 10.0  | 0.1   |
| (*)                      | $\pm$     | 3                     | —                      | —     | 0.0   | 5                     | —                      | —     | 0.3   | 0.3                   | —                      | —     | 0.1   |
| LEUCKART                 | max       | 120                   | —                      | —     | 0.8   | 115                   | —                      | —     | 8.2   | 5                     | —                      | —     | 1.5   |
| <i>Nephtys caeca</i>     | $\bar{x}$ | 0                     | —                      | 0.0   | —     | 0.1                   | —                      | 4.0   | —     | 6                     | —                      | 38.2  | —     |
| FABRICIUS                | $\pm$     | 0                     | —                      | —     | —     | 0.1                   | —                      | —     | —     | 2                     | —                      | —     | —     |
|                          | max       | 0                     | —                      | —     | —     | 3                     | —                      | —     | —     | 85                    | —                      | —     | —     |
| <i>Polydora ligni</i>    | $\bar{x}$ | 52                    | 28                     | 35.7  | 0.1   | 47                    | 25                     | 37.0  | 0.1   | 4                     | 2                      | 13.3  | 0.1   |
| WEBSTER (*)              | $\pm$     | 37                    | 26                     | —     | 0.0   | 25                    | 18                     | —     | 0.0   | 2                     | 1                      | —     | 0.1   |
|                          | max       | 1570                  | 1043                   | —     | 0.8   | 670                   | 479                    | —     | 0.5   | 35                    | 10                     | —     | 2.6   |
| <i>Polydora</i>          | $\bar{x}$ | 12                    | 5                      | 23.8  | 0.0   | 19                    | 5                      | 25.9  | 0.0   | 3                     | 0.2                    | 16.7  | 0.0   |
| <i>ciliata</i> (*)       | $\pm$     | 6                     | 2                      | —     | 0.0   | 8                     | 2                      | —     | 0.0   | 2                     | 0.1                    | —     | 0.0   |
| JOHNSTON                 | max       | 190                   | 59                     | —     | 0.1   | 205                   | 44                     | —     | 0.2   | 55                    | 3                      | —     | 0.3   |

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| Indicators for organic pollution (progressive species of 1st order) | Indicators for slight organic pollution (progr. species of 2nd order) | Adverse indicators (Regressive species) |
|---------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------|
| <i>Capitella capitata</i>                                           | <i>Corophium insidiosum</i>                                           | <i>Bathyporeia sarsi</i>                |
| <i>Nereis diversicolor</i>                                          | <i>Mytilus edulis</i>                                                 | <i>Copidognathus fabriciusi</i>         |
| <i>Polydora ligni</i>                                               | <i>Pygospio elegans</i>                                               | ( <i>Pholoë minuta</i> )                |
|                                                                     | <i>Hydrobia ulvae</i>                                                 | ( <i>Nephtys caeca</i> )                |
|                                                                     | <i>Gammarus salinus</i>                                               | ( <i>Protodrilus chaetifer</i> )        |
|                                                                     | ( <i>Polydora ciliata</i> )                                           | ( <i>Mysella bidentata</i> )            |
|                                                                     | ( <i>Nereis succinea</i> )                                            |                                         |
|                                                                     | ( <i>Idotea baltica</i> )                                             |                                         |
|                                                                     | ( <i>Microdeutopus gryllotalpa</i> )                                  |                                         |
|                                                                     | ( <i>Anaitides mucosa</i> )                                           |                                         |

| Variable                | df | F    | p     |
|-------------------------|----|------|-------|
| Abundance               | 1  | 1.61 | 0.222 |
| Biomass                 | 1  | 0.13 | 0.720 |
| Taxon richness          | 1  | 0.73 | 0.405 |
| Pielou's evenness       | 1  | 2.70 | 0.119 |
| Shannon index           | 1  | 2.29 | 0.149 |
| Trait richness          | 1  | 0    | 0.999 |
| Trait evenness          | 1  | 0.47 | 0.505 |
| Trait dispersion        | 1  | 2.38 | 0.142 |
| Taxonomic composition   |    |      |       |
| treatment at Sampling 1 | 9  | 0.12 | 0.953 |
| treatment at Sampling 2 | 10 | 1.90 | 0.101 |
| Trait composition       |    |      |       |
| treatment at Sampling 1 | 9  | 0.52 | 0.853 |
| treatment at Sampling 2 | 10 | 1.60 | 0.183 |